

SAMMIT POUDYAL

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFILE

Final-year Computer Engineering student exploring full-stack development, data, and AI/ML through hands-on projects. I enjoy learning how different technologies fit together and using them to build practical software. Experienced with React/Next.js, Python, FastAPI, PostgreSQL, and modern AI/ML tools. Seeking **Full Stack** or **AI/ML** internship opportunities.

EDUCATION

Kathmandu University

Bachelor of Engineering in Computer Engineering

Jan 2023-Jan 2027 (Expected)

Dhulikhel, Nepal | CGPA: 3.87 / 4

Relevant Coursework: Data Structures & Algorithms, Neural Networks & Deep Learning, Artificial Intelligence, Software Engineering, Database Management, System Analysis & Design, Computer Architecture, Cloud Computing

SKILLS

Languages: Python, TypeScript, JavaScript, HTML/CSS, C++, C, SQL

Frameworks & Libraries: React.js, Next.js, FastAPI, Tailwind CSS, Node.js

AI/ML: LLM Integration & APIs, RAG, Vector Embeddings, PyTorch, TensorFlow, scikit-learn, LightGBM, NLP, Computer Vision, Sentence-Transformers

Data: NumPy, Pandas, Matplotlib, Feature Engineering

Databases: PostgreSQL (Supabase/pgvector), MongoDB, MySQL, Firebase, SQLite

Tools & DevOps: Git, GitHub, GitHub Actions, Docker, Vercel, Render, Railway, Google Cloud Platform

PROJECTS

- **Skim** | *Python, Next.js, TypeScript, Supabase/pgvector, Groq/Gemini LLM APIs* 2026
 - Built an automated Python ingestion pipeline using feedparser and Sentence-Transformers, scheduled through GitHub Actions to collect, enrich, and curate articles into daily digests.
 - Architected a RAG system combining 384-dimensional MiniLM embeddings with PostgreSQL full-text search, powering source-cited answers through a Gemini/Groq fallback LLM pipeline.
 - Developed a Next.js 16 dashboard with Supabase authentication, pgvector semantic search, admin moderation, and personalized UI preferences; deployed on Vercel.
- **Prometheus** | *Python, LightGBM, FastAPI, React* 2025-2026
 - Built a daily wildfire-risk forecasting system for Nepal using LightGBM and 44 weather, satellite, and terrain features; achieved 0.155 PR-AUC vs. 0.043 baseline across 10-fold cross-validation.
 - Built a FastAPI backend and React/Leaflet interface for district-level risk visualization, what-if weather simulation, and forecast monitoring; containerized with Docker.
- **NepAI** | *React, TypeScript, Zustand, Tailwind CSS* 2026
 - Built an interactive stock-analysis dashboard covering 585 NEPSE-listed stocks, with candlestick charts, technical indicators, and AI-powered price forecasts.
 - Developed portfolio management with real-time profit/loss tracking, protected authentication, stock search, market overviews, and prediction/model-health views.
- **NoteMerge** | *Python, FastAPI, Next.js, MongoDB, Docling/PyMuPDF* 2025
 - Built a full-stack document-processing application that merges PDF/DOCX/PPTX lecture notes into structured, math-rendered study guides using an LLM pipeline with Gemini/Groq fallback, live progress streaming, and failure recovery.

- Implemented Markdown/HTML/PDF export and production deployment across Vercel, Oracle Cloud, and MongoDB Atlas, with rate limiting and restricted CORS.
- **Contrace** | *React, Django, Python, NLP* 2024
 - Built a full-stack content publishing platform using React with functionality for creating and managing written content.
 - Implemented keyword extraction using NLP techniques (TF-IDF and YAKE) to identify key terms for search and recommendations.
- **WorkSpot** | *React, Node.js, APIs* 2023
 - Built a React/Node.js workspace and job-discovery platform with responsive interfaces and REST API integration.

OTHER PROJECTS

- **TamangNetra (तामाङ नेत्र)** | *Python, FastAPI, PyMuPDF, ReportLab, Pandas* 2026 - Google TMT Hackathon
 - Built a multilingual English/Nepali/Tamang document-translation platform using Google TMT API and a coordinate-aware PDF engine with PyMuPDF and ReportLab, preserving document layout, tables, and images.
- **Skill Forge** | *React, FastAPI, PyTorch, scikit-learn, SQLAlchemy* 2026
 - Built a gamified adaptive learning platform with ML-powered learning-style personalization (decision tree + PyTorch MLP) driving quiz difficulty and skill recommendations.
- **Covify 3D** | *Three.js, Tauri, Alpine.js, Spotify API* 2026
 - Built an interactive 3D Spotify visualizer with PKCE OAuth playback controls, packaged as a cross-platform desktop app with Tauri.

LEADERSHIP / EXTRACURRICULAR

- **Track winner, Infinity Hackathon 2026** - [Smart and Secure Future]
- **Logistics Co-Lead**, Hackathon KU IT Meet (2025)
- **Event Manager**, Valorant IT MEET Competition (2024)
- **Logistics Member**, Hackathon event KU IT Meet (2024)
- **Volunteer**, IT Meet (2023)